BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

SECTION – A (Version-A)

Time Allowed: 20 Min | **Maximum Marks:** 18

Note: Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q. Which of these is anatomically incorrect?

(A) The nose has to nasal cavities

(B) The pharynx connects the nasal and oral cavities to the larynx

(C) The larynx contains the vocal cords

(D) The trachea enters the lungs

Q. The electrical potential of cell membrane of neuron when it is not transmitting any signal is called _____

(A) Resting membrane potential

(B) Action potential

(C) Propagation of impulse

(D) Synapse

Q. The ability of an organisms to tolerate dehydration for certain period of time is called.

(A) Hydrophilic

(B) Hydrophobic

(C) Anhydrobiosis

(D) Osmoregulation

Q. Identify the process in which weakened pathogen is introduced into the body

(A) Attenuation

(B) Vaccination

(C) Biological control

(D) Radiotherapy

Q. Which of the following joints can move in only one direction?

(A) Ball and socket joint
(B) Pivot joint
(C) Hing joint
(D) Gliding joint

Q. In the striated muscles, the functional unit of the contractile system is

(A) Z band
(B) Cross bridges
(C) Myofibrils
(D) Sarcomere

Q. Thyroid stimulating hormone is secreted by the endocrine gland named

(A) Pituitary gland
(B) Thyroid gland
(C) Parathyroid gland
(D) Adrenal gland

Q. Fight and flight hormones are produced by:

(A) Gonads
(B) Pituitary
(C) Adrenal medulla
(D) Adrenal cortex

Q. Which one of these is the example of thigmotropism

(A) Plants grow towards light
(B) Root grow downward
(C) Pollen tube grows towards ovule
(D) Vine twines around a tree

Q. The uterine layer which is shed with each monthly cycle is.

(A) Endometrium
(B) Perimetrium
(C) Tunica albuginea
(D) Myometrium

Q. Which one is not a symptom of aging?

(A) Changes in sleeping patterns
(B) Sexual disfunction
(C) A tendency to gain weight decreases
(D) Decreased memory

Q. Genes for Hemophilia are located on

(A) X - chromosome
(B) Y - chromosome
(C) Autosome
(D) Both "X" and "Y" chromosome

Q. "Y" linked traits are inherited

(A) Only by females
(B) Only by males
(C) By both male and female
(D) Only by autosomes

Q. The _____ is a fluid-filled hollow sphere composed of trophoblast and inner cell mass.

(A) Zygote
(B) Morulla
(C) Gastrula
(D) Blastocyst

Q. A fungus penicillium has how many pairs of chromosomes?

(A) 10 pairs
(B) 20 pairs
(C) 02 pairs
(D) One pair

Q. The total number of functional codons are;

(A) 60
(B) 61
(C) 64
(D) 62

Q. Sliding filament hypothesis is currently popular model for the contraction of _____

(A) ATP
(B) Arms
(C) Legs
(D) Muscle fiber

Q. Genetic markers are generally located within

(A) Coding region of the gene
(B) Non-coding region of the gene
(C) Exon region
(D) Both A and C are correct

SECTION – B

Maximum Marks: 40

Note: Q.No 2. Attempt any TEN parts of the following. All parts carry equal marks.

i. Describe the procedure of peritoneal dialysis.

ii. Describe the role of respiratory pigments in human.

iii. Describe the structure and function of inter-vertebral discs.

iv. Describe briefly the sense of smell in human being.

v. Draw a labelled of a nephron.

vi. What is territorial behavior?

vii. Differentiate between learning and innate behavior.

viii. Define linkage and crossing over.

ix. Differentiate between resting membrane potential and active membrane potential.

x. Differentiate between gene mutation and chromosomal mutation.

xi. Differentiate between homologous structures and analogous structures.

xii. State the role of ADH in controlling the proper concentration of urine.

xiii. What is RNA splicing?

SECTION – C

Maximum Marks: 27

Note: Note: Attempt any THREE of the following. All questions carry equal marks

Q3. Q.No:3.

(a) Discuss the major events of human embryonic development during first trimester.

(b) Explain the types of nephrons and why juxtamedullary nephrons are more important in osmoregulation.

Q4. Q.No:4.

Explain the ultra structure of human skeletal muscles. Also draw labelled diagram.

Q5. Q.No:5.

a) Explain Mendel law of segregation.

b) Explain the Genetics of color blindness.

Q6. Q.No:6.

Write a detailed note on any TWO of the following: i. Neo-Darwinism ii. Modern synthesis. iii. Types of Muscles

GROUP – II

SECTION-A (VERSION A)

Time Allowed: 20 minutes | **Maximum Marks:** 18

Note: Q.No.1. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number in provided bubble sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q. The mammalian forebrain is differentiated into the thalamus, limbic system and the:

(A) Cerebellum
(B) Cerebrum
(C) Hippocampus
(D) Hypothalamus

Q. The first major event in organogenesis is _____

(A) Neurulation
(B) Ectoderm formation
(C) Gonadal development
(D) Heart formation

Q. Secretory phase of the menstrual cycle occurs on _____ days.

(A) 15-28
(B) 14th
(C) 6-14
(D) 1-5

Q. In PCR technique primers select _____ of target DNA fragment by annealing with its complementary sequences.

(A) 5' end
(B) 3' end
(C) Both a and b
(D) None of them

Q. Which part of the air passageway possesses cartilage plates in its wall?

(A) Bronchioles
(B) Distal region of bronchi
(C) Proximal region of bronchi
(D) Trachea

Q. Oxytocin is secreted by the endocrine gland named:

(A) Pituitary gland
(B) Thyroid gland
(C) Parathyroid gland
(D) Adrenal gland

Q. How many different kinds of gametes will be formed by an individual who is heterozygous for four gene pairs:

(A) 8
(B) 16
(C) 20
(D) 30

Q. The base of hair follicle is provided with _____ muscle:

(A) Cardiac
(B) Ring
(C) Skeletal
(D) Smooth

Q. The earth's atmosphere contains almost _____ % nitrogen gas:

(A) 40
(B) 60
(C) 80
(D) 81

Q. The most important hormone in initiating and maintaining lactation after birth is:

(A) Estrogen
(B) FSH
(C) Prolactin
(D) Oxytocin

Q. Cell suspension culture of Cinchona ledgeriana produce:

(A) Quinine
(B) Digitoxin
(C) Polludrin
(D) Anti-toxin

Q. The most toxic substances routinely found in the blood are metabolites of what types of molecules?

(A) Proteins
(B) Carbohydrates
(C) Nucleic acids
(D) Fats

Q. Innate behavior is all BUT:

(A) Heritable
(B) Intrinsic
(C) Stereotype
(D) Flexible

Q. Wine is made from the fermentation of _____ by yeast:

(A) alcohol
(B) Glucose
(C) Grape sugar
(D) Maltose

Q. In mitochondria UGA codon act to specify _____ instead stop codon:

(A) Arginine
(B) Valine
(C) Glutamic acid
(D) Tryptophan

Q. Which of the following is NOT recycled in the ecosystem?

(A) Carbon
(B) Sulphur
(C) Energy
(D) Water

Q. Embryo implants in the _____ of the uterus:

(A) Perimetrium
(B) Myometrium
(C) Endometrium
(D) Cervix

Q. Skeletal muscles consist of longest muscle cells and are _____

(A) Mononucleated
(B) Binucleated
(C) Multinucleated
(D) Without nucleus

SECTION-B

Maximum Marks: 40

Note: Q.No.2. Attempt any 10 parts of the following. All parts carry equal marks.

i. Differentiate between sympathetic and parasympathetic nervous system.

ii. What is the importance of cleavage?

iii. What is the role of RNA splicing in post-transcriptional modification?

iv. Write a short note on transgenic bacteria.

v. Differentiate between animal husbandry and livestock management.

vi. What are the types of hormones on the basis of chemical nature?

vii. Write down symptoms of bacterial, viral and mycoplasma pneumonia.

viii. Differentiate between sex limited and sex influenced traits.

ix. Define and give the types of ecological succession.

x. What is cannabis? What are its effects on nervous coordination?

xi. How innate behavior ensures the survival of individuals in a population.

xii. Briefly describe role of ADH as an example of negative feedback.

xiii. Write a short note on accessory glands of human male reproductive system.

SECTION-C

Maximum Marks: 27

Note: Note: Attempt any (THREE) questions of the following. All questions carry equal marks.

Q03. Q.NO.03.

Write a detailed note on evidences of evolution.

Q04. Q.NO.04.

Explain the role and process of polymerase chain reaction (PCR) as a way of gene cloning.

Q05. Q.NO.05.

Explain the ultra structure of human skeletal muscles.

Q06. Q.NO.06.

Write detailed notes of any TWO of the following. a) Breast feeding vs bottle feeding. b) Types of dominance c) Causes of depletion of natural resources